Phase 1: Automated tests

We start by running our test suite via pytest. As you can see, all 36 unit tests pass successfully, confirming that our event bus, simulator, and query indexes are fully functional.

1.1 Execute automated tests:

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\tests> pytest test_agent.py
>>
====================================================================== test session starts ======================================================================
platform win32 -- Python 3.9.10, pytest-8.4.2, pluggy-1.6.0
rootdir: C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\tests
plugins: anyio-4.12.1
collected 36 items

test_agent.py ....................................                                                                                                       [100%]

====================================================================== 36 passed in 2.66s ======================================================================
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\tests> 
fwd-i-search: _
                                   Tejas Shashikant Patil (17 hours ago)   Ln 344, Col 60   Spaces: 4   UTF-8   CRLF   {} Python   venv (3.9.10)
```

Phase 2: Pipeline orchestration and processing

Now we launch the core orchestrator script. This processes our 14-frame security scenario timeline. The event bus automatically routes data through our mock analyzer, running rule evaluations and spitting out real-time alerts for critical events like loitering or fence perimeter breaches.

2.1 Run main ingestion engine:

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent> cd src
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python main.py

═══════════════════════════════════════════════════
  DRONE SECURITY ANALYST AGENT
  Mode: Mock (no API)
═══════════════════════════════════════════════════

Processing 14 frames...

  ◈ Frame 01 [00:01] Main Gate
    → HIGH     | Dark vehicle, person observed at Main Gate at 00:01.
    🚨 ALERT [CRITICAL] [LOITERING] Person loitering at Main Gate, 00:01. Review footage immediately.
    ⚠ ALERT [HIGH] [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:01. Dark vehicle, person o
  ◈ Frame 02 [00:08] Main Gate
    → HIGH     | Person observed at Main Gate at 00:08.
    🚨 ALERT [CRITICAL] [LOITERING] Person loitering at Main Gate, 00:08. Review footage immediately.
    ⚠ ALERT [HIGH] [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:08. Person observed at Mai
  ◈ Frame 03 [02:30] Perimeter Fence - NE Corner
    → CRITICAL | Person observed at Perimeter Fence - NE Corner at 02:30.
    🚨 ALERT [CRITICAL] [INTRUSION CRITICAL] Perimeter fence breach at Perimeter Fence - NE Corner, 02:30. Dispatc
    ⚠ ALERT [HIGH] [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE Corner, 02:30. Pers
    ⚠ ALERT [HIGH] [UNIDENTIFIED] Person without visible ID badge detected at night — Perimeter Fence - NE Co
  ◈ Frame 04 [06:45] Service Entrance
    → LOW      | White truck, person observed at Service Entrance at 06:45.
  ◈ Frame 05 [07:15] Service Entrance
    → LOW      | White truck observed at Service Entrance at 07:15.
  ◈ Frame 06 [08:30] Main Gate
    → LOW      | Blue truck, person observed at Main Gate at 08:30.
  ◈ Frame 07 [09:15] North Parking Lot
    → LOW      | Blue truck, person observed at North Parking Lot at 09:15.
  ◈ Frame 08 [12:00] North Garage
    → LOW      | Blue truck, person observed at North Garage at 12:00.
    ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 3 times today. Last seen at North Gara
  ◈ Frame 09 [12:47] Main Gate
    → LOW      | Blue truck, person observed at Main Gate at 12:47.
    ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 4 times today. Last seen at Main Gate,
  ◈ Frame 10 [14:20] East Courtyard
    → LOW      | Person observed at East Courtyard at 14:20.
  ◈ Frame 11 [17:30] Main Gate
    → LOW      | Blue truck observed at Main Gate at 17:30.
                                   Tejas Shashikant Patil (17 hours ago)   Ln 344, Col 60   Spaces: 4   UTF-
```

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python main.py
  → LOW      | Blue truck, person observed at North Parking Lot at 09:15.
 ◈ Frame 08 [12:00] North Garage
  → LOW      | Blue truck, person observed at North Garage at 12:00.
  ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 3 times today. Last seen at North Gara
 ◈ Frame 09 [12:47] Main Gate
  → LOW      | Blue truck, person observed at Main Gate at 12:47.
  ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 4 times today. Last seen at Main Gate,
 ◈ Frame 10 [14:20] East Courtyard
  → LOW      | Person observed at East Courtyard at 14:20.
 ◈ Frame 11 [17:30] Main Gate
  → LOW      | Blue truck observed at Main Gate at 17:30.
  ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 5 times today. Last seen at Main Gate,
 ◈ Frame 12 [19:30] North Garage
  → LOW      | Blue vehicle, person observed at North Garage at 19:30.
  ⚠ ALERT [HIGH] [LOITERING] Person loitering at North Garage, 19:30. Review footage immediately.
  ⚠ ALERT [HIGH] [ACCESS ATTEMPT] Unauthorized access attempt at North Garage, 19:30.
  ⚠ ALERT [MEDIUM] [REPEAT VEHICLE] 'Blue Ford F150' has been observed 6 times today. Last seen at North Gara
 ◈ Frame 13 [22:10] Loading Dock
  → HIGH     | Dark vehicle, person observed at Loading Dock at 22:10.
  ⚠ ALERT [HIGH] [NIGHT ACTIVITY] Activity detected after hours at Loading Dock, 22:10. Dark vehicle, perso
  ⚠ ALERT [HIGH] [ACCESS ATTEMPT] Unauthorized access attempt at Loading Dock, 22:10.
 ◈ Frame 14 [23:58] Perimeter Fence - NE Corner
  → CRITICAL | Vehicle, person observed at Perimeter Fence - NE Corner at 23:58.
  ⚠ ALERT [HIGH] [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE Corner, 23:58. Vehi

 ──────────────────────────────────────────────────────────────────
  Generating daily intelligence report...

 ══════════════════════════════════════════════════════════════════
  SIMULATION COMPLETE
 ══════════════════════════════════════════════════════════════════

  Event log  → C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\sample_output\event_log.txt
  Alert log  → C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\sample_output\alert_log.txt
  Report     → C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\sample_output\daily_report.txt
  Database   → C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\sample_output\drone_security.db
                                                                                                                python query.py --summary
```

Phase 3: Database Queries

Next, we can query our SQLite database interface. Running the summary flag calculates our total alert aggregates directly from the database tables.

We can also run full-text searches. Looking up 'F150' utilizes SQLite FTS5 extension indexing to instantly grab every timestamp and location where this specific vehicle made an appearance.

3.1 Check the global analytics summary:

```
                                                                                                                    python query.py --summary
>> nv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src>

 ══════════════════════════════════════════════════════════════
  DAILY SECURITY SUMMARY
 ══════════════════════════════════════════════════════════════

  Total alerts fired : 16
  🚨 CRITICAL        : 3
  ⚠ HIGH             : 9
  🔵 MEDIUM          : 4

  Priority Incidents:
    [00:01] Main Gate — [LOITERING] Person loitering at Main Gate, 00:01. Review footage immed
    [00:08] Main Gate — [LOITERING] Person loitering at Main Gate, 00:08. Review footage immed
    [02:30] Perimeter Fence - NE Corner — [INTRUSION CRITICAL] Perimeter fence breach at Perimeter Fence - NE Co
    [00:01] Main Gate — [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:01. Da
    [00:08] Main Gate — [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:08. Pe
    [02:30] Perimeter Fence - NE Corner — [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE
    [02:30] Perimeter Fence - NE Corner — [UNIDENTIFIED] Person without visible ID badge detected at night — Per
    [19:30] North Garage — [LOITERING] Person loitering at North Garage, 19:30. Review footage im
    [19:30] North Garage — [ACCESS ATTEMPT] Unauthorized access attempt at North Garage, 19:30.
    [22:10] Loading Dock — [NIGHT ACTIVITY] Activity detected after hours at Loading Dock, 22:10.
    [22:10] Loading Dock — [ACCESS ATTEMPT] Unauthorized access attempt at Loading Dock, 22:10.
    [23:58] Perimeter Fence - NE Corner — [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE

  Repeat Vehicles (1):
    Blue Ford F150 — seen 6x at 08:30, 09:15, 12:00, 12:47, 17:30, 19:30
```

3.2 Search for the specific suspicious truck:

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python query.py --object "F150"
>>

────────────────────────────────────────────────────
  FTS Search: 'F150' — 6 frame(s) found
────────────────────────────────────────────────────

  Frame 06 [08:30] Main Gate
     Risk: LOW
  Summary: Blue truck, person observed at Main Gate at 08:30.

  Frame 07 [09:15] North Parking Lot
     Risk: LOW
  Summary: Blue truck, person observed at North Parking Lot at 09:15.

  Frame 08 [12:00] North Garage
     Risk: LOW
  Summary: Blue truck, person observed at North Garage at 12:00.

  Frame 09 [12:47] Main Gate
     Risk: LOW
  Summary: Blue truck, person observed at Main Gate at 12:47.

  Frame 11 [17:30] Main Gate
     Risk: LOW
  Summary: Blue truck observed at Main Gate at 17:30.

  Frame 12 [19:30] North Garage
     Risk: LOW
  Summary: Blue vehicle, person observed at North Garage at 19:30.
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python query.py --alerts HIGH
>>
```

3.3 Filter by high-risk alerts:

(we filter the alerts table to surface only high-severity threats, giving our physical security response team a structured target window list)

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python query.py --alerts HIGH
>>

────────────────────────────────────────────────────
  Alerts ≥ HIGH — 12 alert(s)
────────────────────────────────────────────────────

  🚨 [CRITICAL] 00:01 — Main Gate
  Rule: LoiteringRule
  [LOITERING] Person loitering at Main Gate, 00:01. Review footage immediately.

  ⚠️ [HIGH] 00:01 — Main Gate
  Rule: NightTimeActivityRule
  [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:01. Dark vehicle, person observed at Main Gate at 00:01.

  🚨 [CRITICAL] 00:08 — Main Gate
  Rule: LoiteringRule
  [LOITERING] Person loitering at Main Gate, 00:08. Review footage immediately.

  ⚠️ [HIGH] 00:08 — Main Gate
  Rule: NightTimeActivityRule
  [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:08. Person observed at Main Gate at 00:08.

  🚨 [CRITICAL] 02:30 — Perimeter Fence - NE Corner
  Rule: FenceIntrusionRule
  [INTRUSION CRITICAL] Perimeter fence breach at Perimeter Fence - NE Corner, 02:30. Dispatch security immediately.

  ⚠️ [HIGH] 02:30 — Perimeter Fence - NE Corner
  Rule: NightTimeActivityRule
  [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE Corner, 02:30. Person observed at Perimeter Fence - NE Corner at 02:30.

  ⚠️ [HIGH] 02:30 — Perimeter Fence - NE Corner
  Rule: UnidentifiedNightPersonRule
  [UNIDENTIFIED] Person without visible ID badge detected at night — Perimeter Fence - NE Corner, 02:30.

  ⚠️ [HIGH] 19:30 — North Garage
  Rule: LoiteringRule
  [LOITERING] Person loitering at North Garage, 19:30. Review footage immediately.
```

3.4 Filter by time:

```
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src> python query.py --time "22:00" "06:00"
>>

────────────────────────────────────────────
  Time Range: 22:00 – 06:00 — 5 frame(s)
────────────────────────────────────────────

  [00:01] Main Gate
  ⚠ HIGH — Dark vehicle, person observed at Main Gate at 00:01.

  [00:08] Main Gate
  ⚠ HIGH — Person observed at Main Gate at 00:08.

  [02:30] Perimeter Fence - NE Corner
  🚨 CRITICAL — Person observed at Perimeter Fence - NE Corner at 02:30.

  [22:10] Loading Dock
  ⚠ HIGH — Dark vehicle, person observed at Loading Dock at 22:10.

  [23:58] Perimeter Fence - NE Corner
  🚨 CRITICAL — Vehicle, person observed at Perimeter Fence - NE Corner at 23:58.
(venv) PS C:\Users\Tejas\Documents\Course Material\my projects (MIT)\drone-security-agent\drone-security-agent\src>
```

Phase 4: Output logs

Once the simulation completes, it generates our static assets in the sample_output folder. This includes a detailed per-frame security log, an alert log tracking every anomaly, and our structured text intelligence report briefing.

```
alert_log.txt M ×
sample_output > alert_log.txt
=== DRONE SECURITY ANALYST — ALERT LOG ===

[00:01] [CRITICAL] Main Gate | LoiteringRule
  → [LOITERING] Person loitering at Main Gate, 00:01. Review footage immediately.
[00:01] [HIGH    ] Main Gate | NightTimeActivityRule
  → [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:01. Dark vehicle, person observed at Main Gate at 00:01.
[00:08] [CRITICAL] Main Gate | LoiteringRule
  → [LOITERING] Person loitering at Main Gate, 00:08. Review footage immediately.
[00:08] [HIGH    ] Main Gate | NightTimeActivityRule
  → [NIGHT ACTIVITY] Activity detected after hours at Main Gate, 00:08. Person observed at Main Gate at 00:08.
[02:30] [CRITICAL] Perimeter Fence - NE Corner | FenceIntrusionRule
  → [INTRUSION CRITICAL] Perimeter fence breach at Perimeter Fence - NE Corner, 02:30. Dispatch security immediately.
[02:30] [HIGH    ] Perimeter Fence - NE Corner | NightTimeActivityRule
  → [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE Corner, 02:30. Person observed at Perimeter Fence - NE Corner at 0
[02:30] [HIGH    ] Perimeter Fence - NE Corner | UnidentifiedNightPersonRule
  → [UNIDENTIFIED] Person without visible ID badge detected at night — Perimeter Fence - NE Corner, 02:30.
[12:00] [MEDIUM  ] North Garage | RepeatVehicleRule
  → [REPEAT VEHICLE] 'Blue Ford F150' has been observed 3 times today. Last seen at North Garage, 12:00. Verify vehicle authorization.
[12:47] [MEDIUM  ] Main Gate | RepeatVehicleRule
  → [REPEAT VEHICLE] 'Blue Ford F150' has been observed 4 times today. Last seen at Main Gate, 12:47. Verify vehicle authorization.
[17:30] [MEDIUM  ] Main Gate | RepeatVehicleRule
  → [REPEAT VEHICLE] 'Blue Ford F150' has been observed 5 times today. Last seen at Main Gate, 17:30. Verify vehicle authorization.
[19:30] [HIGH    ] North Garage | LoiteringRule
  → [LOITERING] Person loitering at North Garage, 19:30. Review footage immediately.
[19:30] [HIGH    ] North Garage | AfterHoursAccessRule
  → [ACCESS ATTEMPT] Unauthorized access attempt at North Garage, 19:30.
[19:30] [MEDIUM  ] North Garage | RepeatVehicleRule
  → [REPEAT VEHICLE] 'Blue Ford F150' has been observed 6 times today. Last seen at North Garage, 19:30. Verify vehicle authorization.
[22:10] [HIGH    ] Loading Dock | NightTimeActivityRule
  → [NIGHT ACTIVITY] Activity detected after hours at Loading Dock, 22:10. Dark vehicle, person observed at Loading Dock at 22:10.
[22:10] [HIGH    ] Loading Dock | AfterHoursAccessRule
  → [ACCESS ATTEMPT] Unauthorized access attempt at Loading Dock, 22:10.
[23:58] [HIGH    ] Perimeter Fence - NE Corner | NightTimeActivityRule
  → [NIGHT ACTIVITY] Activity detected after hours at Perimeter Fence - NE Corner, 23:58. Vehicle, person observed at Perimeter Fence - NE Co
```